National University of Singapore

Bridging Mathematics

Semester 1, AY2025/2026

Time allowed: 1 hour and 30 minutes

Mock Test

INSTRUCTIONS TO CANDIDATES

(a) Write down your name clearly in the space provided above.

(b) Only this booklet will be collected at the end of the examination. Do not insert any loose pages in this booklet.

(c) This paper contains **ten (10)** questions and comprises **twenty-two (22)** printed pages. Answer **ALL** questions to the best of your ability.

(d) This is a CLOSED BOOK examination. All calculations and working must be presented and laid out systematically.

(e) Unless otherwise stated, all numerical answers must be given exactly.

FOR EXAMINER USE ONLY

1	2	3	4	5	6	7	8	9	10

List of Commonly Used Probability Models

Discrete Random Variables

- **Uniform Distribution** $X \sim U(n)$

  $$P(X = x) = \frac{1}{n}, \quad \mathbb{E}[X] = \frac{1}{n}\sum x, \quad \mathrm{Var}(X) = \frac{1}{n}\sum (x - \mathbb{E}[X])^2.$$

- **Geometric Distribution** $X \sim \mathrm{Geom}(p)$

  $$P(X = x) = (1-p)^{x-1} p, \quad \mathbb{E}[X] = \frac{1}{p}, \quad \mathrm{Var}(X) = \frac{1-p}{p^2}.$$

- **Binomial Distribution** $X \sim \mathrm{Binom}(n, p)$

  $$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad \mathbb{E}[X] = np, \quad \mathrm{Var}(X) = np(1-p).$$

- **Poisson Distribution** $X \sim \mathrm{Pois}(\lambda)$

  $$P(X = x) = e^{-\lambda}\frac{\lambda^x}{x!}, \quad \mathbb{E}[X] = \lambda, \quad \mathrm{Var}(X) = \lambda.$$

Continuous Random Variables

- **Uniform Distribution** $X \sim U(a, b)$

  $$p(x) = \begin{cases} \frac{1}{b-a} & \text{if } a < x < b \\ 0 & \text{otherwise} \end{cases}, \quad \mathbb{E}[X] = \frac{a+b}{2}, \quad \mathrm{Var}(X) = \frac{(b-a)^2}{12}.$$

- **Exponential Distribution** $X \sim \mathrm{Exp}(\lambda)$

  $$p(x) = \begin{cases} \lambda e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise} \end{cases}, \quad \mathbb{E}[X] = \frac{1}{\lambda}, \quad \mathrm{Var}(X) = \frac{1}{\lambda^2}.$$

Question 1 [10 marks]

Determine whether each of the following statements are true or false. Justify your answers.

(a) There exists a linear system with exactly three solutions.

(b) If $\mathbf{A}$ is a symmetric matrix, then $\mathbf{A}^2$ must be a symmetric matrix.

(c) If $\mathbf{A}$ is an invertible $n \times n$ matrix, then the homogeneous system $\mathbf{A}\vec{x} = \vec{0}$ has infinitely many solutions.

(d) The function $T : \mathbb{R}^n \to \mathbb{R}^n$ given by $T(\vec{x}) = \vec{x} + \vec{b}$ is a linear transformation for all $\vec{b} \in \mathbb{R}^n$.

(e) You flip a fair coin 100 times. The probability of obtaining exactly 50 heads is greater than the probability of obtaining exactly 25 heads.

Question 2 [10 marks]

In his *Mathematical Classic* (張邱建算經), the 5th Century Chinese mathematician Zhang Qiujian poses the famous *Hundred Fowl Problem*:

> 今有雞翁一直錢五；雞母一直錢三，雞雛三直錢一。凡百錢買雞百隻。問雞翁、 母、雛各幾何?
>
> A rooster costs five coins, a hen costs three coins, and three chicks cost one coin. With 100 coins, we buy 100 of them. How many roosters, hens, and chicks can we buy?

Show your work to Question 2 in the space below.

You may use this page to continue your work on Question 2.

Question 3 [10 marks]

Find all matrices $\mathbf{X}$ that satisfy the following equation:

$$\mathbf{X}\begin{pmatrix} -1 & -1 & 0 \\ 0 & -1 & 0 \\ 5 & 3 & 2 \\ -1 & -1 & -2 \end{pmatrix} = \begin{pmatrix} 5 & 3 & 2 \\ 0 & -1 & 2 \end{pmatrix}.$$

Show your work to Question 3 in the space below.

You may use this page to continue your work on Question 3.

Question 4 [10 marks]

Consider the following system of linear equations:

$$\begin{cases} x + ay + bz = 1 \\ x + y + bz = 2 \\ x + y + (a-b)\,z = -1 \end{cases}.$$

Determine the conditions on a and b such that:

(a) the linear system has no solution;

(b) the linear system has a unique solution, and write down the unique solution;

(c) the linear system has a infinitely many solutions in one parameter, and write down this solution;

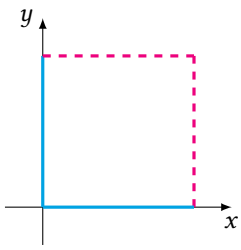
(d) the linear system has a infinitely many solutions in two parameters, and write down this solution.

Show your work to Question 3 in the space below.

You may use this page to continue your work on Question 3.

Question 5 [10 marks]

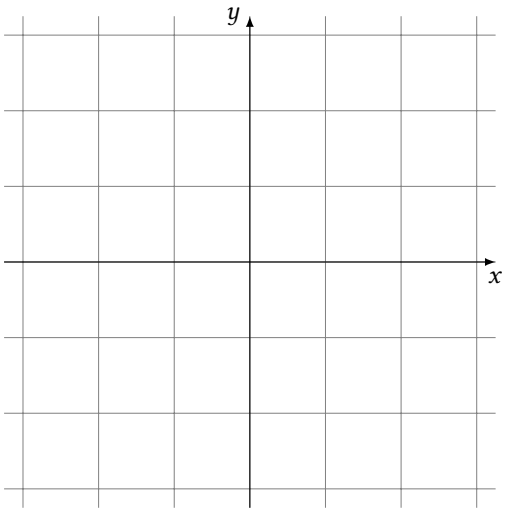
Consider the unit square in $\mathbb{R}^2$ with vertices at $(0,0)$, $(1,0)$, $(0,1)$, and $(1,1)$. Note that the sides along the axes are drawn with filled lines, whereas the remaining edges are drawn with dashed lines.



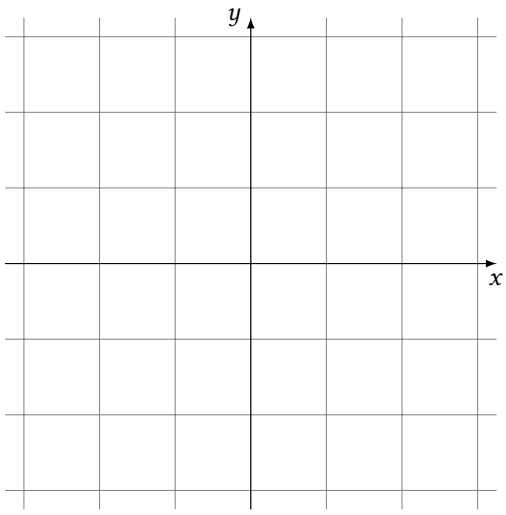
On each set of axes below, draw the image of the square under the linear transformation described by the given matrix.

Show your work to Question 5 in the space below.

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$$



$$\mathbf{B} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}$$



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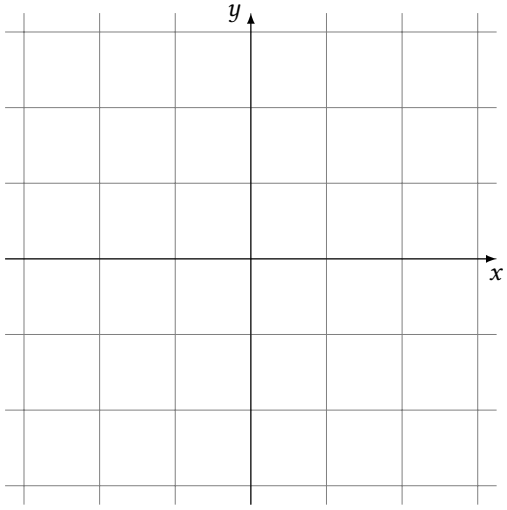
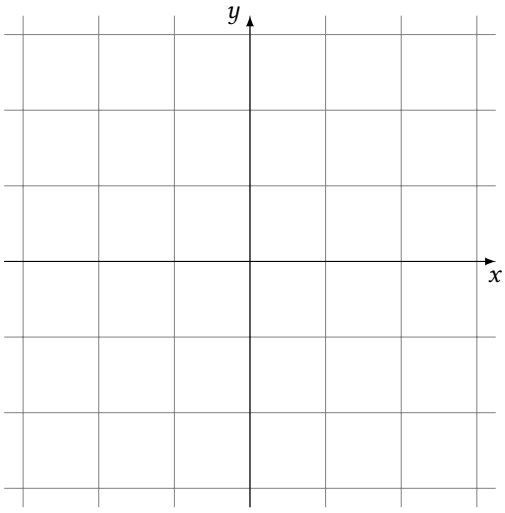
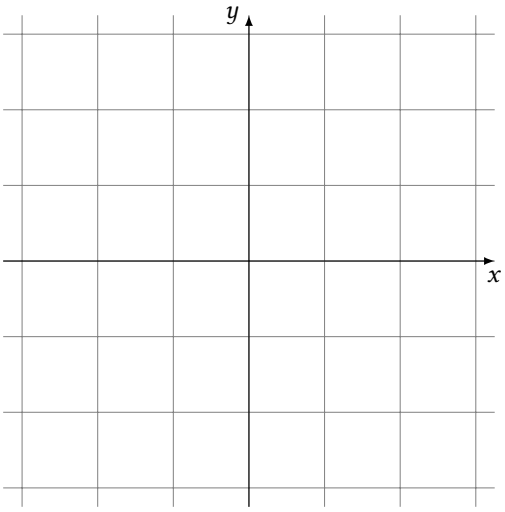
$$\mathbf{C} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$



$$\mathbf{D} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

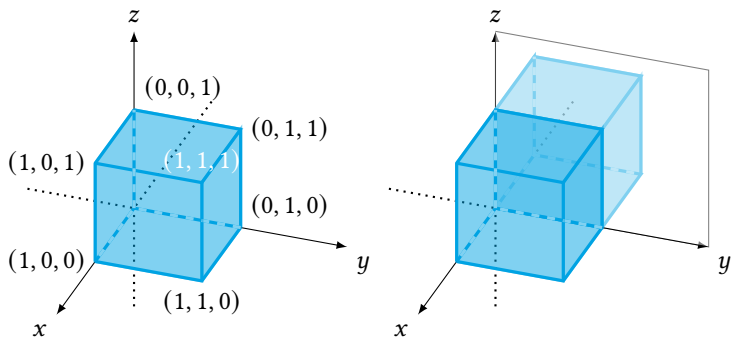


$$\mathbf{E} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$



Question 6 [10 marks]

Consider a unit cube situated in the first octant (with x, y, z all positive) of three-dimensional space, as shown below.



If we place a mirror along the yz-plane, the cube gets reflected into another cube in the second octant (with x negative and y, z positive).

(a) Let T represent the transformation the cube undergoes as it is reflected through the mirror on the yz-plane. For each of the cube's vertices, write down the coordinates of their reflection in the mirror.

(b) Hence, or otherwise, find the standard matrix $\mathbf{T}$ of this transformation.

(c) Is the matrix $\mathbf{T}$ invertible? If so, state $\mathbf{T}^{-1}$ and explain the geometric significance of your answer; otherwise, explain why $\mathbf{T}^{-1}$ does not exist.

Show your work to Question 6 in the space below.

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You may use this page to continue your work on Question 6.

Question 7 [10 marks]

In an election, 40% of all voters voted for Party A, 49% voted for Party B, and 11% voted for Party C. Among those who voted for Party A, 30% were female; among those who voted for Party B, 50% were female, and among those who voted for Party C, 70% were female.

(a) Find the probability that the voter is a female.

(b) If a voter is female, what is the probability that she voted for Party A?

(c) If a voter is female, what is the probability that she voted for Party B or Party C?

Show your work to Question 7 in the space below.

You may use this page to continue your work on Question 7.

Question 8 [10 marks]

Let $X \sim \text{Geom}(p)$ be a geometrically distributed random variable.

(a) Write down the possible values of X.

(b) The geometric distribution is *memoryless*. Briefly explain what this means.

(c) Suppose that $p = 0.2$. Compute $\mathbb{E}(X \mid X \geq 2)$.

Show your work to Question 8 in the space below.

You may use this page to continue your work on Question 8.

Question 9 [10 marks]

A line segment of unit length is cut into two pieces at a randomly chosen point along the segment. Let X be difference in the lengths between the two resulting segments.

(a) Find the probability that $X < 0.2$.

(b) Find the expected value and the variance of X.

Show your work to Question 9 in the space below.

You may use this page to continue your work on Question 9.

Question 10 [10 marks]

Let X be the time (in hours) before a fully charged mobile device runs out of battery. Suppose that the probability density function of X is given by

$$p(x) = \begin{cases} cx^{-2} & x > 10, \\ 0 & x \leq 10. \end{cases}$$

(a) Find the value of c. Hence, find the cumuluative distribution function of X.

(b) Find the probability that a fully charged device will last more than 20 hours.

(c) Suppose you have six such devices. What is the probability that at least three of them will function for at least 15 hours? State any assumptions.

Show your work to Question 10 in the space below.

You may use this page to continue your work on Question 10.

You may use this page for your rough work.

You've made it to the end!